ANN Machine Failure Prediction - Results Summary

Dense(6, ReLU) -> Dense(4, ReLU) -> Dropout(0.2) -> Dense(1, Sigmoid) | Adam optimizer | Binary cross-entropy

1. Dataset

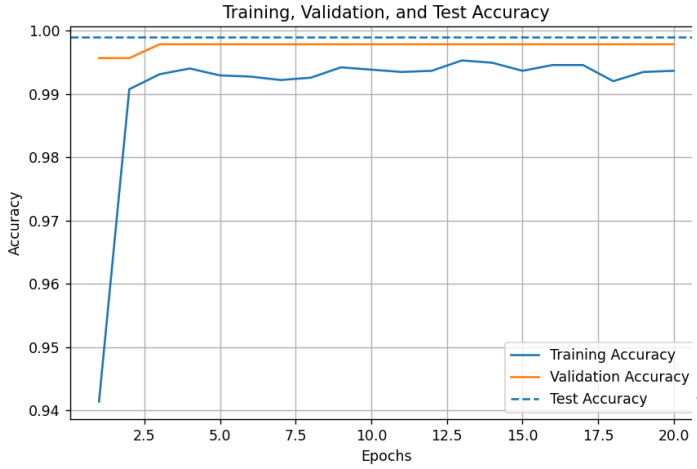
Item	Value
Samples	9,815
Target	Machine failure
Classes	0 = no failure, 1 = failure
Inputs	Speed, torque, tool wear, TWF, HDF, PWF, OSF
Mean speed	1526.38 rpm
Mean torque	40.33 Nm
Mean wear	107.91 min
Failure rate	2.91%

2. Training

The network was trained for 20 epochs. Training accuracy improved from 94.14% to 99.36%. Validation accuracy quickly reached 99.78%, while validation loss decreased strongly in the early epochs.

3. Test Performance

Metric	Result
Test accuracy	99.90%
Test loss	0.0101
Val. accuracy (epoch 20)	99.78%
Val. loss (epoch 20)	0.0261



4. Confusion Matrix and Classification Results

	Pred. 0	Pred. 1
Actual 0	2880	0
Actual 1	8	57

Class 1 performance: precision = 1.00, recall = 0.88, F1-score = 0.93. The model detected 57 failures and missed 8 failures.

5. Interpretation

- The model converged quickly; most accuracy improvement happened in the first few epochs.
- Validation accuracy is almost constant because most validation samples were already classified correctly.
- Loss can still decrease while accuracy stays constant because the model becomes more confident.
- The dataset is highly imbalanced, so accuracy alone is not enough.
- For failure prediction, missed failures are more important than false alarms.

6. Conclusion

Overall, the ANN gives excellent validation and test accuracy. However, because failures are rare, the confusion matrix and failure-class recall should be emphasized. The model is reliable for normal cases and detects most failures, but recall could be improved to reduce missed failure cases.